

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
Applied Business Analytics

Task – 3

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Task (on house_price_regression_dataset.csv) (Don't include in print document)

Q.1 Find the correlation between Year_Built and House_Price.

code:

```
import pandas as pd
df = pd.read_csv("house_price_regression_dataset - house_price_regression_dataset.csv")
correlation = df["Year_Built"].corr(df["House_Price"])
print("Correlation between Year_Built and House_Price:", correlation)
```

Output:

```
Correlation between Year_Built and House_Price: 0.051967361931064486
```

Q.2 Create a correlation matrix for all numerical columns.

Code:

```
import pandas as pd
df = pd.read_csv("house_price_regression_dataset - house_price_regression_dataset.csv")
correlation_matrix = df.corr(numeric_only=True)
print("Correlation Matrix:")
print(correlation_matrix)
```

Output:

```
Correlation Matrix:
      Square_Footage  Num_Bedrooms  Num_Bathrooms  Year_Built \
Square_Footage      1.000000      -0.043564      -0.031584      -0.022392
Num_Bedrooms       -0.043564      1.000000      0.022848      -0.015820
Num_Bathrooms       -0.031584      0.022848      1.000000      -0.021063
Year_Built          -0.022392      -0.015820      -0.021063      1.000000
Lot_Size            0.089479      -0.009355      0.034923      -0.061050
Garage_Size         0.030593      0.113761      0.024846      -0.025485
Neighborhood_Quality -0.008357      -0.049024      0.017585      -0.009549
House_Price         0.991261      0.014633      -0.001862      0.051967

      Lot_Size  Garage_Size  Neighborhood_Quality  House_Price
Square_Footage  0.089479    0.030593             -0.008357    0.991261
Num_Bedrooms   -0.009355    0.113761             -0.049024    0.014633
Num_Bathrooms   0.034923    0.024846              0.017585   -0.001862
Year_Built     -0.061050   -0.025485             -0.009549    0.051967
Lot_Size        1.000000    0.002436              0.037630    0.160412
Garage_Size     0.002436    1.000000             -0.011287    0.052133
Neighborhood_Quality 0.037630   -0.011287             1.000000   -0.007770
House_Price     0.160412    0.052133             -0.007770    1.000000
```

Q.3 Perform linear regression using:

X = Year_Built

Y = House_Price

Display the coefficient and intercept.

Code:

```
import pandas as pd
from sklearn.linear_model import LinearRegression
df = pd.read_csv("house_price_regression_dataset - house_price_regression_dataset.csv")
X = df[["Year_Built"]]
Y = df[["House_Price"]]
model = LinearRegression()
model.fit(X, Y)
print("Coefficient:", model.coef_[0])
print("Intercept:", model.intercept_)
```

Output:

```
... Coefficient: 638.6524884727302
Intercept: -649854.0823295021
```

Q.4 Calculate the R2 score for the regression model:

Year_Built - House_Price

Code:

```
import pandas as pd
from sklearn.linear_model import LinearRegression
from sklearn.metrics import r2_score
df = pd.read_csv("house_price_regression_dataset - house_price_regression_dataset.csv")
X = df[["Year_Built"]]
Y = df[["House_Price"]]
model = LinearRegression()
model.fit(X, Y)
Y_pred = model.predict(X)
r2 = r2_score(Y, Y_pred)
print("R2 Score:", r2)
```

Output:

```
... R2 Score: 0.0027006067060743044
```

Q.5 Calculate the R2 score for:

Square_Footage - House_Price

Code:

```
import pandas as pd
from sklearn.linear_model import LinearRegression
from sklearn.metrics import r2_score
df = pd.read_csv("house_price_regression_dataset - house_price_regression_dataset.csv")
X = df[["Square_Footage"]]
Y = df[["House_Price"]]
model = LinearRegression()
model.fit(X, Y)
Y_pred = model.predict(X)
r2 = r2_score(Y, Y_pred)
print("R2 Score:", r2)
```

Output:

```
... R2 Score: 0.9825992634147099
```

Q.6 Calculate the R2 score for:

Neighborhood_Quality - House_Price

Code:

```
import pandas as pd
from sklearn.linear_model import LinearRegression
from sklearn.metrics import r2_score
df = pd.read_csv("house_price_regression_dataset - house_price_regression_dataset.csv")
X = df[["Neighborhood_Quality"]]
Y = df["House_Price"]
model = LinearRegression()
model.fit(X, Y)
Y_pred = model.predict(X)
r2 = r2_score(Y, Y_pred)
print("R2 Score:", r2)
```

Output:

```
... R2 Score: 6.037338916142776e-05
```

Q.7 Compare the R2 scores of all three regression models.

Regression Model	R ² Score	Percentage Explained
Year_Built → House_Price	0.0027	0.27%
Square_Footage → House_Price	0.9826	98.26%
Neighborhood_Quality → House_Price	0.0000604	0.01%